



Microsoft® ADO.NET

Presented by

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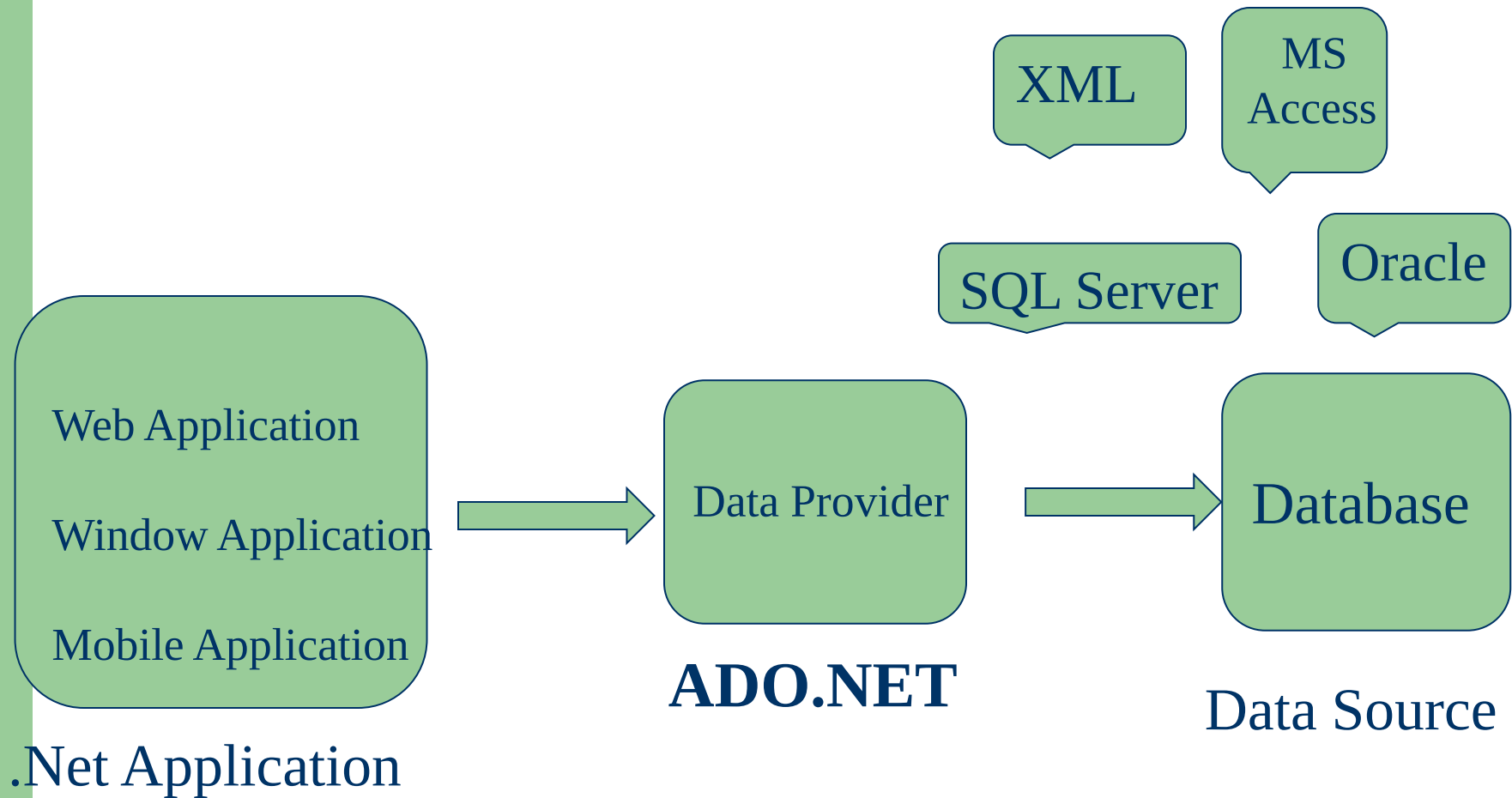
Agenda

- **ADO.NET Overview**
- ADO.NET Architecture
- Data Provider
- DataReader
- DataAdapter Object
- DataSet
- DataObject
- Data Reader VS Data Set
- ADO VS ADO.NET

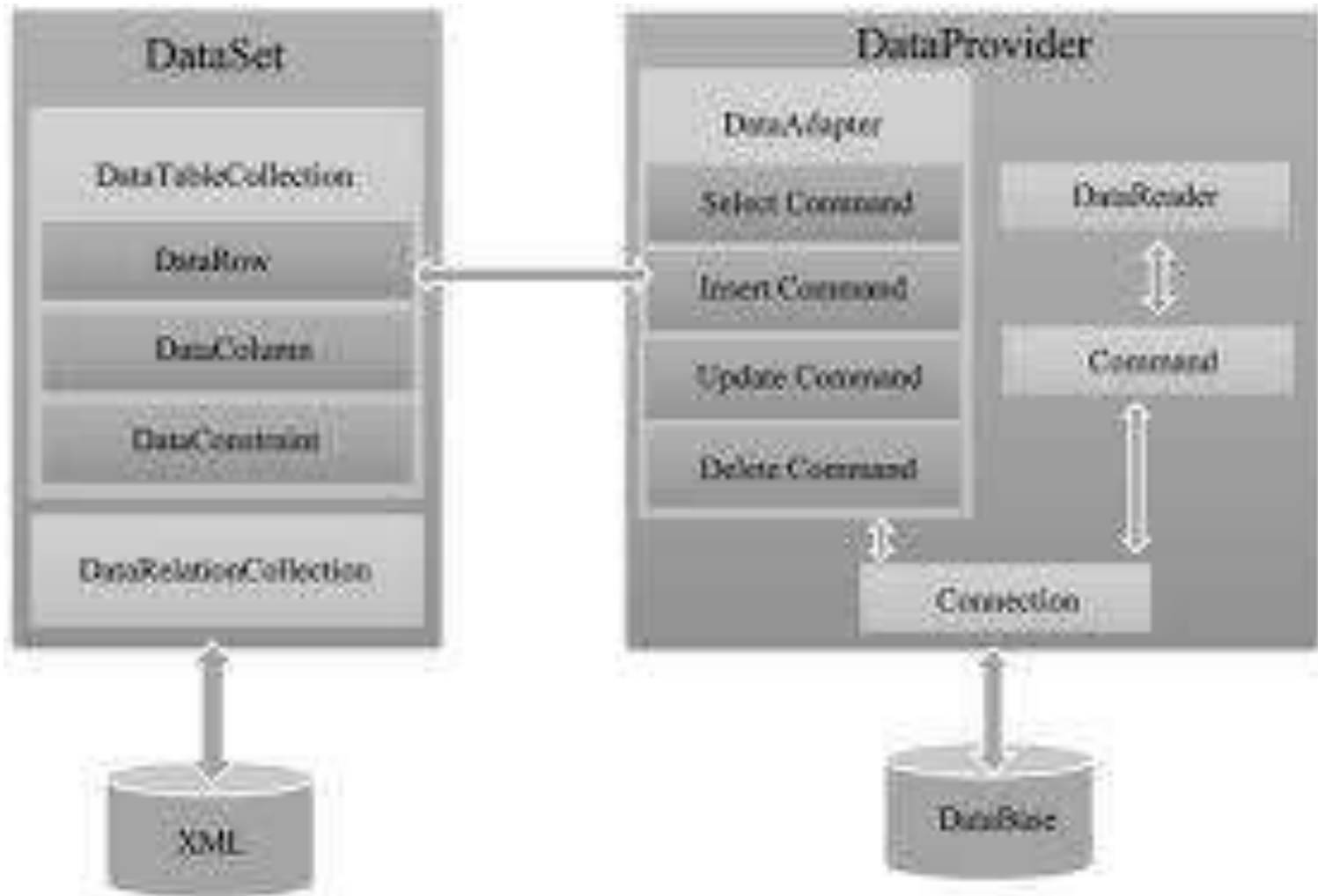
INTRODUCTION OF ADO.NET

- ADO Stand for **Active-X Data Object**.
- Microsoft ADO.Net is the latest improvement after ADO.net provider platform interoperability and scalable data access.
- .Net framework data is transmitted in the **Extensible Markup Language**(XML).
- ADO.Net technology used for working with data and database of all types. It provides access to data source as microsoft sql server and to data source exposed through OLEDB and XML.

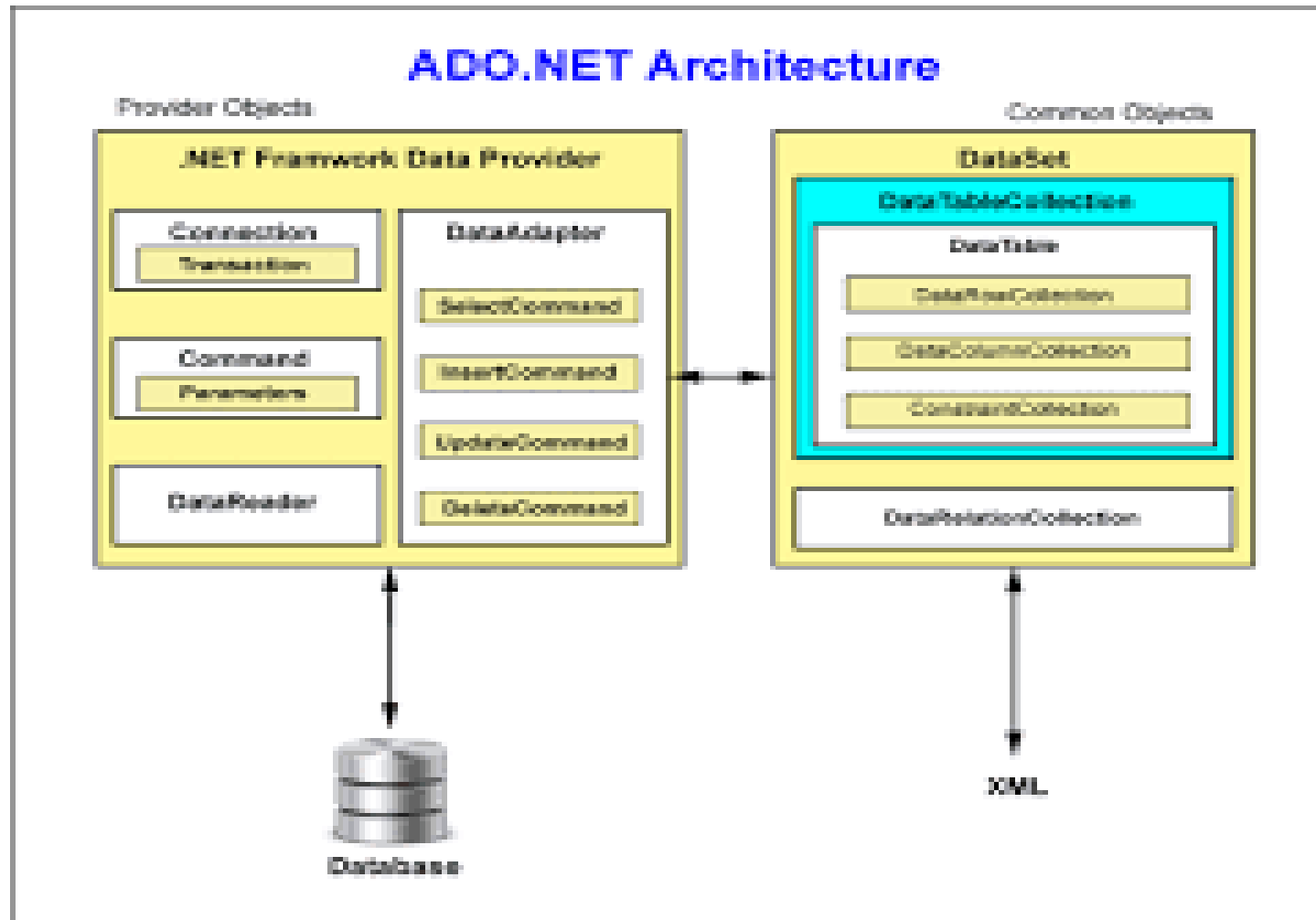
WHAT IS ADO.NET?



ADO.NET Architecture Diagram



ADO.NET Architecture Diagram



Data Source

Data Provider	Namespace	DataBases
SQLClient	System.Data.SqlClient	SQLServer
OracleClient	System.Data.OracleClient	Oracle
OLEDB	System.Data.OleDb	MS Aceess
ODBC	System.Data.Odbc	Configured Databases

Data Provider

- The connection object all of the information required to open a channel to the database in the connection string property.
- The .net application can not directly execute our SQL code. It only understands C#. If a .net application need to retrieve data or to do some insert, update, and delete operations from or to a database.
- Database Connection
- Prepare SQL Command
- Execute the Command
- Retrieve the results and display the web application

Data Reader

- Provides methods and properties that deliver a forward-only stream of data rows from a data source
- When a DataReader is used, parts of the ADO.NET model are cut out, providing faster and more efficient data access.
- It provides a way to fetch data row by row from a data source, making it efficient for scenarios where you only need to retrieve and display data (not update it).
- Very **fast** and **lightweight**.
- Best for scenarios where you just need to **read and display data**.

Data Adapter

- Provides a set of methods and properties to retrieve and save data between a DataSet and its source data store
- Allows the use of stored procedures
- Connects to the database to fill the DataSet and also update the database

Data Set

- Data table represents a single table.
- Data set represents multiple tables as a data table objects.
- Data table and data set resides in dis-connected architecture.
- Represents a cache of data that contains tables, columns, relationships, and constraints, just like a database.
- It is collection of data tables that contain the data.
- Data set tabular representation of data. Tabular representation means it represents data into row and column format.
- We can use data set in combination with data adapter class. The data set contains the copy of the data we requested.
- A data set local copy of your database table that get populated in client pc.

Introduction about SQL Server

- **SQL Server** is a **Relational Database Management System (RDBMS)** developed by **Microsoft**.
- It is designed to store, retrieve, and manage data efficiently, and it supports **Structured Query Language (SQL)** as its primary query language.
- SQL Server is widely used in enterprise environments for handling small to very large databases, offering strong security, reliability, and integration with Microsoft tools.
- SQL Server is a powerful, secure, and scalable RDBMS by Microsoft that helps organizations store, process, and analyze data effectively.

Connection with SQL Server

- In the **Connect to Server** dialog box, enter -> **Server type** → Database Engine – Server name: localhost.
- `string connectionString = "Server=localhost;
Database=MyDatabase; Integrated Security=True;";`

```
SqlConnection conn = new
SqlConnection(connectionString);

    try {          conn.Open();
        MessageBox.Show("✓ Connection Successful!");    }
catch (Exception ex) {
    MessageBox.Show("✗ Error: " + ex.Message);
    } finally
    {
        conn.Close();
    }
```

Data Grid View

- The **DataGridView** is one of the most useful controls in **Windows Applications** for **working with tabular data**.
- The DataGridView control provides a customizable table for displaying data.
- The DataGridView class allows customization of cells, rows, columns, and borders through the use of properties such as `DefaultCellStyle`, `ColumnHeadersDefaultCellStyle`, `CellBorderStyle`, and `GridColor`.
- The DataGridView control supports the standard Windows Forms data binding model, so it can bind to a variety of data sources.

ExecuteReader()

- The ADO.NET method **ExecuteReader()** is used with a Command object (like SqlCommand) to fetch data from a database in a forward-only, read-only manner. It returns a DataReader object (e.g., SqlDataReader when using SQL Server).
- Returns a **DataReader** object.
- Used for **SELECT queries**.
- Reads data **row by row** (forward-only).
- Requires the database **connection to remain open** while reading.

ExecuteNonQuery()

- This method is commonly used for update, insert, or Delete statements, where the only returned value is the number of records affected.
- This method can be return results if you call a stored procedure that has output parameters.
- ExecuteNonQuery() is used with the sqlCommand to execute SQL statements that do not return rows of data.
- Returns an **integer** → number of rows affected. Used for **action queries**.

ExecuteScalar()

- ExecuteScalar() is used when you want to retrieve a single value from the database.
- It executes the query and returns the **first column of the first row** in the result set.
- Any extra rows or columns are **ignored**.
- Commonly used with aggregate functions like COUNT(), SUM(), MAX(), etc
- ExecuteScalar(single value) → returns **object**.

Data Reader

- Data Reader is a connection oriented service.
- Data Reader is used to retrieve **Read only** (can not update) and **forward only** (can not read backward / Random) data from a database.
- Data Reader fetches the records from database and stored in the network buffer and given whenever requests.
- Data Reader is like a forward only record set. It fetches one rows at a time so very less network cost compare to DATASET.
- As a one row at a time is stored in memory in data reader it increase application performance and reduced system overheads.

Data Set

- Data Set Class in ado.net operates in a entirely disconnected nature.
- Data set is a **collection of in-memory table**. It can be used with multiple data source.
- Data Set releases the data after loading all the data in memory.
- Dataset fetches all the rows at a time. i.e. it fetches all data from the data source at a time to its memory area.
- There is more system overheads in dataset as it fetches all the data from the data source at a time in memory.

ADO	ADO.NET
➤ Classic ado regulars active connection with the data store.	➤ ADO.NET architecture works while the data store is disconnected.
➤ Data is stored in binary format	➤ Data is stored in xml.
➤ XML integration is not possible.	➤ XML integration is possible.
➤ It uses the object named Record set to reference data from the data store.	➤ It uses data set object for data access and representation.
➤ Firewall might prevent execution of classic ADO.	➤ ADO.NET has firewall proof and its execution will never be interrupted.
➤ Classic ADO architecture includes client side curser and server side curser.	➤ ADO.NET architecture doesn't include such cursors.
➤ We cannot send multiple transactions using a single connection instance.	➤ We can send multiple transactions using single connection instance.

Command Object

- Information submitted to a database as a query via a Connection object
- Two provider-specific classes
 - SqlCommand
 - OleDbCommand
- Input and output parameters are supported, along with return values as part of the command syntax
- Results are returned in the form of streams. Accessed by:
 - DataReader object
 - DataSet object via a DataAdapter

Connection object

- Connects to databases.
- Two provider-specific classes
 - SqlConnection
 - OleDbConnection.
- Connections can be opened in two ways:
 - Explicitly by calling the Open method on the connection
 - Implicitly when using a DataAdapter.
- Connections handle transactions

ADO.NET Namespaces

System.data	Core namespace, defines types that represent data
System.Data.Common	Types shared between managed providers
System.Data.OleDb	Types that allow connection to OLE DB compliant data sources
System.Data.SqlClient	Types that are optimized to connect to Microsoft® SQL Server
System.Data.SqlTypes	Native data types in Microsoft® SQL Server

Importing the ADO.NET Namespaces

Needed to build a data access application

- **For OLE DB:**

- Imports System.Data

- Imports System.Data.OleDb

- **For SQL Server:**

- Imports System.Data

- Imports System.Data.SqlClient

Introduction to Crystal Reports and its integration with Visual C#

- **Crystal Reports** is a powerful **reporting tool** developed by SAP. It allows developers to design and generate professional-looking reports from various data sources (like SQL Server, Oracle, MySQL, XML, etc.) (System applications & Products in data Processing).
- It is widely used in business applications because of its ability to create detailed, formatted, and interactive reports.
- Crystal Reports is a business intelligence application used to design and generate reports from various data sources.
- It allows users to create customizable reports with features like grouping, summarizing, and parameter linking. These reports can then be exported to various formats or printed.
- **Integration with Visual C#**
- Crystal Reports can be integrated into a C# Windows or Web application to generate and display reports dynamically.
- Crystal Reports integrates with Visual C# within the Visual Studio environment, enabling developers to incorporate reporting functionalities into their applications.

- **Install Crystal Reports for Visual Studio:-** Download and install SAP Crystal Reports Developer Edition for Visual Studio. It adds the Crystal Report viewer control and necessary libraries.
- **Create a New Report:** In your project → Right-click → Add → New Item → Crystal Report. Use the Report Wizard to connect to your database and choose fields.
- **Integration with Visual C#**
- **Design the Report:-** Drag and drop fields, apply grouping, sorting, and formatting. Insert charts, formulas, or images if needed. Add CrystalReportViewer to Form. Open your Windows Form → Drag and drop CrystalReportViewer from the toolbox. Resize it to fit the form.
- **Adding a Report to a Project:** This initiates a wizard-driven process to define the report's structure and data source.
- **Data Source Connection:** Crystal Reports can connect to various data sources, including databases like SQL Server, Oracle, MySQL, and even .NET DataSets. When integrating with C#.

- **Report Design:** Crystal Report Designer in Visual Studio provides a graphical interface to design the report layout data source onto the report, adding text objects for headings, and incorporating visual elements.
- **Displaying Reports with CrystalReportViewer:** This control is used. This control is dragged onto a form and then configured to load and display the designed Crystal Report file.
- **Dynamic Report Generation:** This includes setting report parameters, assigning data sources at runtime (e.g., based on user input or specific queries), and handling events related to the report viewer.
- **Exporting and Printing:** Crystal Reports provides programmatic options within C# to export reports to different file formats (like PDF, Excel) or send them directly to a printer.

Designing reports and working with data sources c#

- **Microsoft SQL Server Reporting Services (SSRS):** A robust, enterprise-level reporting platform for creating and managing reports, often used with SQL Server databases.
- **Databases:** SQL Server, MySQL, Oracle, etc., accessed via `adp.net` (`SqlConnection`, `SqlCommand`, `SqlDataReader`, `SqlDataAdapter`) or Object-Relational Mappers (ORMs) like Entity Framework.
- **Datasets:** In-memory representations of data, often populated from databases or other sources. Visual Studio provides a Dataset Designer for creating typed datasets.
- **Drag-and-Drop Interface:** Most reporting tools provide a visual designer where you can drag and drop report items (tables, text boxes, images, charts) onto the report canvas.
- **Data Binding:** Connect report items to specific data fields from your chosen data source.
- **Security:** Manage data source credentials securely and control access to sensitive report data.
- **Crystal Report Designer** to drag-and-drop fields, text objects, tables, charts, etc Format fonts, colors, and layout.

Formatting

- **Formatting in Crystal Reports using C#** Formatting is the part where you control **fonts, colors, alignment, borders, conditional styles, and number/date formats** inside your report.
- **For Ex.**
- **Font / Size / Style:** Right-click a field → Format Object → Font tab.
- **Alignment:** Right-click a field → Format Object → Paragraph tab → Left, Right, Center.
- **Borders & Background:** Border tab → apply lines or shading.
- **Number / Date Formatting:** Right-click → Format Field → Choose custom formats (currency, percentage, yyyy-MM-dd, etc.).
- **Suppress / Hide Fields:** Right-click field → Suppress (No Drill-Down).

Grouping

- **Grouping** is one of the most powerful features in Crystal Reports because it allows you to **organize records by a field** (like Department, Category, or Date) and then show summaries (like totals, counts, averages) for each group.
- **Grouping** organizes records by a specific field.
- **Example:** If you have an **Employees** table, you can group by **Department** so that all employees in "HR" appear together, then "IT", etc. **Group Header** (title for the group) **Detail Section** (actual records) **Group Footer** (totals/summary for that group).
- Open your .rpt report in Visual Studio → Go to **Insert** → **Group**. Choose the field you want to group by (e.g. **Employees, Department**). → **Group Header Section** → **Group Footer Section**.
- **Adding Summaries:** - Right-click a numeric field (e.g., salary) → Insert → Summary → Choose Sum, Average, Count, etc. Place it in the Group Footer (to show total per department).

Adding parameters

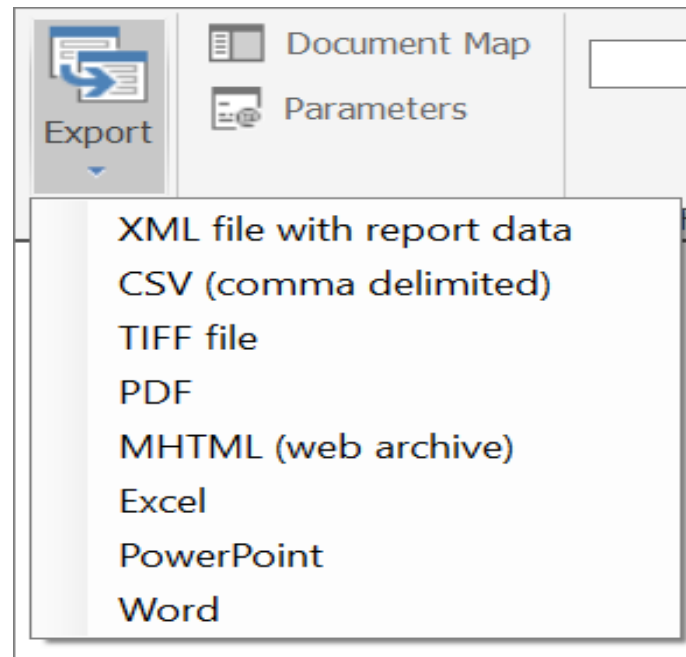
- **Crystal Reports** parameters are used when you want to filter data, pass values dynamically, or allow the user to choose input before the report runs.
- **Open Your Report:-** Open the .rpt file in crystal reports designer.
- **Create a New Parameter:** Go to the Field Explorer window Right-click Parameter Fields → New → Name your parameter → Choose Value Type → (Optional) Provide a Prompt Text → Define Default/Value list if you want to limit options
- **Use the Parameter in Your Report:** In Record Selection Formula → Go to Report → Selection Formulas → Record... In Report Header/Footer : Drag the parameter field onto the report so users can see what was selected.
- **Running the Report:** When the report runs, a prompt dialog will appear asking the user to enter/select a value for the parameter. If multiple parameters exist, Crystal will prompt for each.
- **Passing Parameters from C# (if needed)** If you're using Crystal Reports in a C# Windows Forms app, you can pass parameters.

Summarizing report data

- Open your .rpt file in the Crystal Reports Designer.
- Right-click on the field you want to summarize (ex. Salary).
- Choose **Insert** → **Summary**.
- Select the type of summary (Sum, Average, Count, Minimum, Maximum).
- The summary in **Group Footer** (to summarize per group) or **Report Footer** (to summarize the entire report).

Exporting and deploying reports in different formats

- Reporting Services provides many rendering extensions, supporting exports of reports to common file formats.
- The rendering extensions support file formats with soft breaks (for example, Word or Excel), hard-page breaks (for example, PDF or TIFF), or data only (for example, CSV or Atom compliant XML).



- You want to know how to **export Crystal Reports into different formats (PDF, Excel, Word, etc.)** and then **deploy** them in a C# application.
- Crystal Reports provides the ExportToDisk and exportTostream methods.
- Ex. Export Word
- `rpt.ExportToDisk(ExportFormatType.WordForWindows, @"C:\Exports\SalesReport.doc");`